

STADIOcare 30-Day Patient Readmission Capstone Project

To further investigate and understand the main factors associated with 30-day patient readmissions, the STADIOcare organisation needs to provide relevant data that directly addresses the problem statement. The requested data should provide sufficient information to analyse patient characteristics, admission details, clinical history, follow-up care and readmission outcomes.

Patient and Admission Data

Attributes	Column name	Data type	Example	Additional information
Patient identifier	patient_id	String (Nominal)	P001245	Anonymised unique identifier for each patient.
Admission identifier	admission_id	String (Nominal)	ADM009821	Unique identifier for each hospital admission.
Patient age	age	Integer (Discrete)	68	Age of the patient at admission.
Admission type	admission_type	String (Nominal)	Emergency	Indicates whether the admission was emergency, elective/planned, or another applicable category.
Admission score	admission_source	String (Nominal)	Emergency Department	Indicates where the patient came from before admission.
Admission date	admission_date	Datetime	2025-03-14 10:32	Date and time of admission.
Discharge date	discharge_date	Datetime	2025-03-19 14:15	Date and time of admission.
Length of time in hospital	time_in_hospital	Float (Continuous)	6.8	Number of days/nights the patient stayed in hospital.
ward	ward	String (Nominal)	Ward 29	Ward in which the patient was treated.
Number of procedures	num_procedures	Integer (Discrete)	5	Number of procedures performed during the admission.
Number of diagnoses	num_diagnoses	Integer (Discrete)	4	Number of diagnoses recorded during the admission.
Discharge status	discharge_status	String (Nominal)	home	Indicates the patient's discharge outcome or disposition.

Patient Medical History and Healthcare Utilization

Attributes	Column name	Data type	Example	Additional information
Gender	gender	String (Nominal)	Male	Gender of the patient
Race	race	String (Nominal)	African	Patient's recorded racial category.
Medical specialty	medical_specialty	String (Nominal)	Cardiology	Medical specialty associated with the patient's care.
Number of medications	number_medications	Integer (Discrete)	7	Number of medications recorded during the admission.
Previous inpatient encounters	number_inpatient	Integer (Discrete)	5	Number of previous inpatient encounters for the patient.
Previous emergency encounters	number_emergency	Integer (Discrete)	3	Number of previous emergency encounters for the patient.
Previous outpatient encounters	number_outpatient	Integer (Discrete)	6	Number of previous outpatients encounters for the patient.

Readmission and Follow up Data

Attributes	Column name	Data type	Example	Additional information
Readmitted within 30 days	readmitted_30_days	Boolean	TRUE	Shows if patient was readmitted within 30 days of discharge.
Readmission date	readmission_date	Datetime	2025-04-02 11:20	Date and time of the patient's readmission.
Days to readmission	days_to_readmission	Integer (Discrete)	16	Number of days between discharge and readmission.
Readmission type	readmission_type	String (Nominal)	Emergency	Shows the type of readmission.
Follow-up required	follow_up_required	Boolean	TRUE	Shows whether follow-up care was required after discharge.
Follow-up appointment attended	follow_up_attended	Boolean	FALSE	Shows whether the patient attended the scheduled follow-up appointment.
Days to follow-up	days_to_follow_up	Integer (Discrete)	5	Number of days between discharge and the scheduled follow-up appointment.

